

Low-Cost Open-Source Absorbance Spectrophotometry on a Monochrome Global-Shutter Camera

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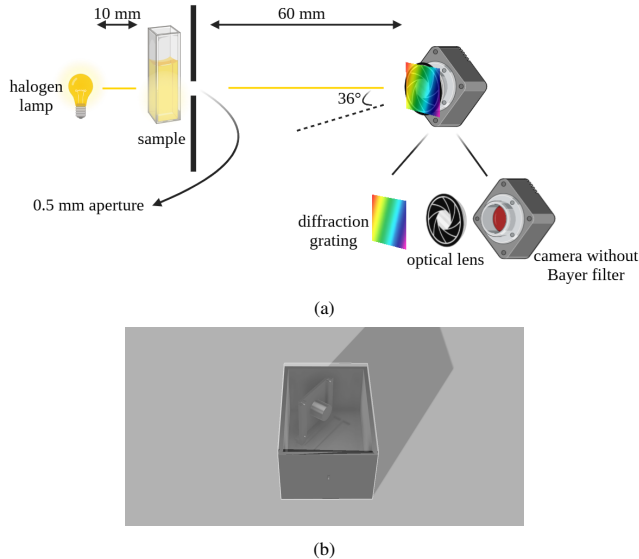


Figure 1. (a) Optical path. Light from the halogen source crosses the cuvette and the entrance slit and is dispersed by a diffraction grating bonded directly to the front face of the lens. The sensor is tilted by 36° so that the first order of the visible band falls across it. (b) The assembled instrument. [replace with the hero photograph of the built unit](#)

ABSTRACT

We present MonoSpectro, an open-source visible-range absorbance spectrophotometer built from a monochrome global-shutter camera and a 1000 mm^{-1} diffraction grating bonded directly to the lens, for about €166 in components, together with a calibration transfer that makes its absorbance readings quantitatively comparable with those of a commercial benchtop instrument. The correction expresses the deviation between the two instruments as a wavelength-dependent gain and offset, expands both to first degree, and fits them in a single least-squares pass over every wavelength and every calibration sample — four coefficients for the whole spectrum. Fitted on nine food colourings and evaluated once on four laboratory dyes measured in a later session, it reduces the root-mean-square error against the reference by 72 % over 420 nm to 780 nm, and improves every sample. Fitting the same relation independently at every wavelength, with 180 times as many coefficients, recovers only 54 %: with few calibration samples, constraining the model is what makes it generalise. All hardware designs, firmware and analysis code are released under the MIT licence.

Index Terms— Calibration transfer, absorbance spectroscopy, CMOS image sensor, low-cost instrumentation, open-source hardware, water quality monitoring.

I. INTRODUCTION

Absorbance spectrophotometry is a staple measurement across chemistry, biology and environmental monitoring, but laboratory-grade accuracy arrives bundled with cost, bulk and mains power. That rules it out precisely where sustained measurement is most valuable: unat-

tended and in the field, on a buoy or a moored station [1–3]. A substantial literature of low-cost and open-source photometers has grown in response [4–6], and consumer image sensors in particular have become credible detectors for quantitative work, in purpose-built spectrometers [7–9] and in smartphone photometry validated on real analytes [10–13]. MonoSpectro builds directly on that line of work.

What is less often demonstrated is that the numbers such an instrument produces are quantitatively comparable with those of a commercial instrument *on samples its calibration never saw*. Many published validations fit and evaluate on the same set of solutions [7, 9, 12], which measures how well a model reproduces its own training data rather than whether it holds on new chemistry. Where readings are to be pooled across instruments and sites, that is the property that matters [14, 15].

In analytical chemistry this is the calibration-transfer problem, studied for decades in near-infrared spectroscopy [16–18]: direct and piecewise direct standardisation map spectra recorded on one instrument onto those of another [19, 20], and slope-and-bias schemes do the same with far fewer degrees of freedom [21], all of them normally fitted on tens of transfer standards. This letter asks what transfers when there are nine. We model the deviation as a wavelength-dependent gain and offset, constrain both to a first-degree expansion, and test the result once on dyes measured in a later session, comparing not against the uncorrected instrument but against a freer version of the same model.

II. THE PLATFORM

The optical train is deliberately minimal: a 10 W halogen lamp, a standard 10 mm cuvette, a 0.5 mm entrance slit, and the wide-angle M12 lens supplied with the camera, carrying a 1000 mm^{-1} transmission diffraction grating film bonded to its front surface. A collimator is not indispensable in this geometry: the slit sits 60 mm from the lens (Fig. 1a), far enough that the beam reaching the grating is close to parallel. For a transmission grating of period d the first-order maxima satisfy

$$m\lambda = d(\sin \theta_i + \sin \theta_m). \quad (1)$$

Equation (1) fixes the two geometric choices visible in Fig. 1a. At normal incidence it places the first order between 23.6° at 400 nm and 53.1° at 800 nm, so the visible band arrives spread over 29.5° and centred near 38° : the camera assembly is therefore tilted 36° from the slit axis, which brings that band onto the sensor instead of past its edge. The same spread sets the requirement on the lens, which must accept at least 29.5° of field; the stock wide-angle lens, specified at 110° horizontally, does so with room to spare and its short focal length is what keeps the whole first order inside a 1/4-inch sensor.

The detector is a 1 Mpixel monochrome global-shutter sensor read out at 1280×720 . Monochrome matters: with no Bayer mosaic every pixel is an unfiltered intensity reading, and no colour interpolation stands between the light and the numbers [22]. It is read at 8 bits, so exposure must bring the brightest part of the blank close to saturation without reaching it, and it carries no infrared-cut filter.

Acquisition runs on a single-board computer (SBC), a Raspberry Pi 4. Frames are captured through the picamera2 library; every col-

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Table 1. Summary of components.

Item	Specification
Camera	Arducam OV9281 — 1 MP mono, global shutter, NoIR, MIPI
Lens	M12 wide angle, 130° diagonal, as supplied with the camera
Grating	Edmund Optics #4621, 25 400 lines/inch ($\approx 1000 \text{ mm}^{-1}$), transmission film
Light source	Osram Halostar Starlite 64415 S 12 V, 10 W, G4, 2800 K
Cuvette	Plastic, 10 mm path length
Compute	Raspberry Pi 4 (SBC)
Enclosure	3D printed, opaque filament
Total cost	$\approx \text{€}166^*$

*Components only; the reference instrument is not included.

umn of a user-defined region of interest is averaged vertically to give one intensity value per pixel column; a low-order polynomial maps pixel position to wavelength, following the procedure described in [23]; and absorbance is reported against a stored blank as

$$A = \log_{10}(I_0/I), \quad (2)$$

where I_0 is the blank intensity and I that of the sample at the same wavelength. The interface is served over HTTP, and the Pi can raise its own wireless access point, so the instrument needs no infrastructure in the field. Table 1 lists the components and Fig. 1b shows the assembled unit.

All reference measurements were made on a K LAB Optizen Alpha double-beam UV-Vis spectrophotometer: Czerny–Turner monochromator with a 1200 mm^{-1} blazed grating, tungsten-halogen and deuterium sources, silicon photodiode detection, 1 nm spectral bandwidth over 190 nm to 1100 nm, wavelength accuracy $\pm 0.3 \text{ nm}$, and photometric accuracy ± 0.002 at 0.5 absorbance units. It plays the part of the master instrument in the sense of the standardisation literature [18, 19]: the quantity MonoSpectro is asked to reproduce is what this instrument reads.

III. CALIBRATION TRANSFER

The instrument’s absorbance readings deviate from the reference in a way that varies with wavelength but is stable across samples. That deviation arises from the spectral response of the detector, the transmission of the optics and the stray-light floor of the enclosure [22, 24], none of which depend on what is in the cuvette, so it is a property of the *instrument* rather than of the samples used to measure it. We therefore model it directly, as a linear relation at each wavelength,

$$A_{\text{ref}}(\lambda) = a(\lambda)A_{\text{dev}}(\lambda) + b(\lambda). \quad (3)$$

What matters is not (3) itself, which is fixed, but how much freedom $a(\lambda)$ and $b(\lambda)$ are given when they are estimated. Estimating them independently at each wavelength and smoothing the resulting curves afterwards is the established approach, and the one direct standardisation takes [19, 20]. Over the working range, sampled at 1 nm, that is two coefficients at each of 361 wavelengths, 722 in all, estimated from nine samples.

The alternative adopted here constrains both curves to first-degree polynomials in wavelength and fits them in a single least-squares pass that uses every wavelength and every training sample at once, so smoothness belongs to the model instead of being a filter applied afterwards. A first-degree fit makes $a(\lambda)$ and $b(\lambda)$ straight lines and the whole correction four numbers. The degree is set rather than searched because the optics give no reason for either curve to be complicated: what they have to capture is a gentle drift with wavelength. Searching confirms it from the other side — higher degrees scored marginally better in cross-validation, which rewards flexibility, and then did worse on the test set. The wavelength range is still chosen by leave-one-out on the training set alone. The same form of correction is used for the camera platform in [23], where it is fitted and reported

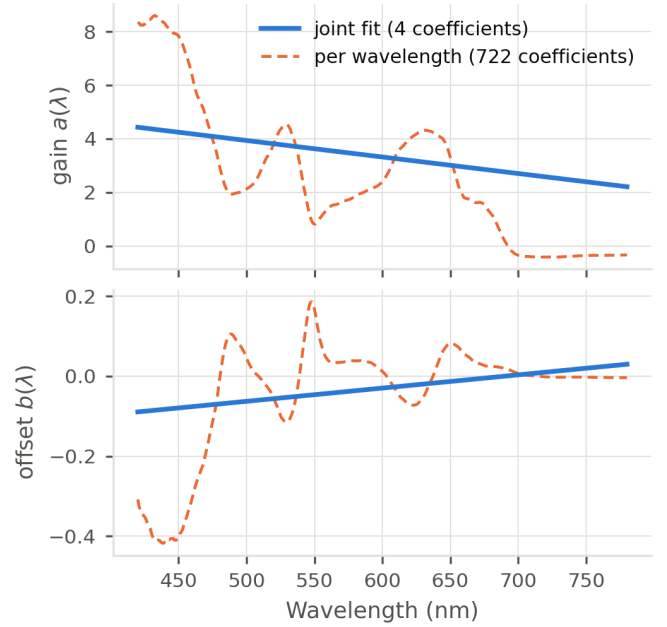


Figure 2. The fitted correction. The per-wavelength fit is free to follow the training samples, while the first-degree joint fit is two straight lines. The constrained model keeps the drift and drops the excursions.

over a different sample set. Fig. 2 shows what the two versions look like once fitted.

Three further steps belong to the data processing rather than to the model. Both sides are passed through a Savitzky–Golay filter [25] before the correction is fitted or applied, because the correction is pointwise and noise is amplified wherever $a(\lambda) > 1$; a window swept between 25 and 85 nm puts the error minimum between 45 and 55 nm, and 45 nm — the least smoothing that reaches the plateau — is used. Since (2) has a hard floor at zero, the corrected output is constrained to be non-negative, without which the fitted line drifts below zero in the near-baseline regions where both the gain and the training data are small and noisy. That constraint is imposed as a rounded floor rather than a clip: over a transition of fixed half-width ε — a processing constant, not one of the four fitted coefficients — the output is blended into zero by a quadratic matching both value and slope at each join. Scored on the near-zero region where it acts, error falls from 0.024 at a hard clip to 0.021 at $\varepsilon = 0.15$.

IV. EXPERIMENTAL PROTOCOL

The training set comprised nine aqueous solutions prepared from a commercial twelve-colour concentrated liquid food-colouring set (6 mL per bottle; Amazon ASIN B08X6FMHKP), each made by diluting one drop of dye in 10 mL of distilled water, which spreads absorbance across the visible band, much as organic dyes are used to characterise low-cost photometers elsewhere [12]. The manufacturer discloses neither the identity nor the concentration of the colorants, so they are named here by their commercial colour names and their reference spectra are given in Fig. 3.

The test set comprised four laboratory-grade dyes, each prepared in a 100 mL volumetric flask with distilled water as the solvent: methylene blue (1.1 mg), rhodamine B (1.4 mg), bromothymol blue (4.5 mg, measured in its acid form) and congo red (1.6 mg), giving 11, 14, 45 and 16 mg L^{-1} respectively. These solutions were prepared and measured in a session later than the training set; both sessions used the same optical configuration and the same procedure, so the test set is separated from the training set by session and by chemistry alone. Every choice in the correction — degree, wavelength range, smoothing window — was fixed on the training set before these four dyes were processed, and they were scored once against the reference; no result below was used to revise the model.

Each reported spectrum is the mean of ten consecutive readings of a single preparation, taken in plastic cuvettes of 10 mm path length, at a stable ambient temperature of 25°C and after a five-minute lamp warm-up on both instruments. The reference instrument is double-

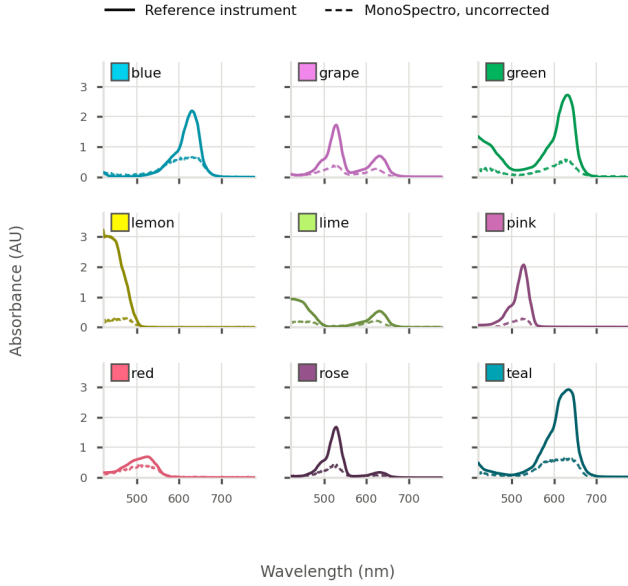


Figure 3. The nine commercial food colourings used to fit the correction, as recorded by the reference instrument and by MonoSpectro before any correction. Each swatch is the solution’s own transmitted colour. The device tracks the position and width of every band but reads systematically low on the absorbance scale, which is the deviation the response function models.

Table 2. Test-set performance. RMSE is in absorbance units, per sample and averaged. R^2 is pooled over all four samples and all 361 wavelengths, that is 1444 points, and N_c is the number of fitted coefficients.

	Uncorrected	Per wavelength	Joint, 1st degree
Bromothymol blue	0.440	0.099	0.149
Congo red	0.440	0.180	0.099
Rhodamine B	0.389	0.233	0.064
Methylene blue	0.116	0.129	0.073
Mean RMSE	0.346	0.160	0.096
Pooled R^2	0.149	0.825	0.936
N_c	0	722	4

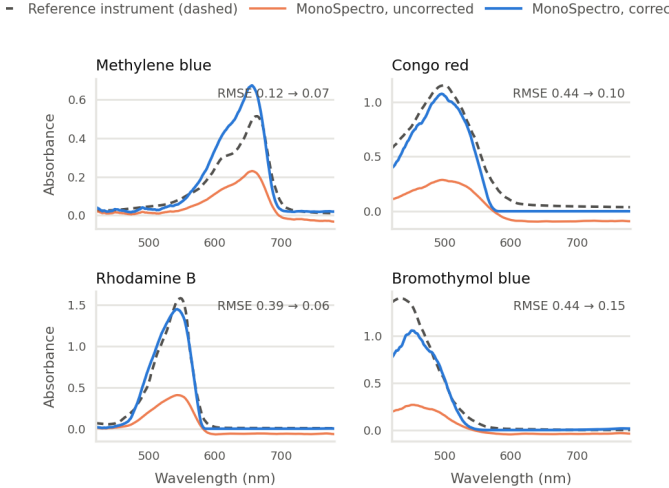


Figure 4. The test set: four laboratory dyes measured in a later session, shown uncorrected, corrected, and as recorded by the reference instrument (dashed).

beam and so carries its blank continuously; on MonoSpectro the same blank of distilled water, in the same cuvette, was re-taken between samples. Agreement is quantified as the root-mean-square error (RMSE) between the device and reference spectra over 420 nm to 780 nm, and is reported throughout beside the do-nothing baseline of applying no correction at all.

V. RESULTS AND DISCUSSION

Table 2 gives the test-set results and Fig. 4 shows the corresponding spectra. The correction reduces the mean RMSE from 0.346 to 0.096 absorbance units, a reduction of 72 %, and every sample improves, by between 37 % and 84 %. The three processing steps contribute in sequence: input smoothing alone brings the mean to roughly 0.17,

the non-negativity constraint to 0.099, and the rounded floor together with the wider smoothing window to 0.096.

Fitting gain and offset independently at every wavelength leaves a mean RMSE of 0.160 and a pooled R^2 of 0.825; the joint first-degree fit, with four coefficients instead of 722, reaches 0.096 and 0.936. Constraining the model improves generalisation rather than costing it. The per-wavelength version has enough freedom to track the idiosyncrasies of nine particular samples, and it was this that degraded methylene blue in earlier versions of the correction: that dye already agreed reasonably well with the reference beforehand, and a freely flexing wavelength curve pulled it toward the average behaviour of the training set. The same reasoning explains why leave-one-out selection of the degree was set aside — on nine samples cross-validation rewarded flexibility that then failed on unseen chemistry.

Independently of the absorbance scale, the spectral shape the instrument records tracks the reference closely: fitting a single linear factor per sample gives coefficients of determination of 0.98, 0.95, 0.92 and 0.89 across the four test dyes, and the recorded band maxima fall within ± 18 , -8 , $+1$ and -5 nm of the reference positions. Placing bands at the right wavelengths with the right widths is the difficult part of building a spectrometer; the absorbance scale is what the response function corrects.

The corrected spectra are nonetheless not laboratory-grade, and Fig. 4 should be read with that in mind: a mean error of 0.096 absorbance units is some fifty times the photometric accuracy quoted for the reference. Bromothymol blue is the worst case — the one dye the constrained fit improves less than the per-wavelength fit does, still well short of the reference below 500 nm — and methylene blue is overshot at its peak. Three known effects contribute, none of them characterised. The grating relation (1) is non-linear in pixel position, the wide-angle lens distorts most at the sensor edges where the blue end falls, and the tilted sensor gives a magnification that varies across the spectrum; a rational function in pixel position would describe that combined geometry better than the polynomial now used, and replacing it is the most direct improvement available. Stray light is the second: the grating is bonded to the lens and the path inside the enclosure is open, with no baffles or order-sorting filter, so light from one part of the spectrum reaches pixels assigned to another — the classic cause of compressed absorbance at high optical density [24, 26] — and the finite spectral bandwidth acts the same way on narrow bands [27]. The third is precision, which is unquantified: each spectrum is the mean of ten readings of one preparation, so repeatability across preparations cannot be given here. This is a portable instrument costing two orders of magnitude less than the reference, and the results support a narrow claim — that the absorbance scale transfers well enough to make readings comparable — rather than a metrological characterisation.

VI. CONCLUSION

A linear response function with wavelength-dependent gain and offset, expanded to first degree and fitted jointly as four coefficients, transfers the absorbance scale of a low-cost camera spectrophotometer onto that of a commercial benchtop spectrophotometer for chemistry it has never seen, reducing test-set error by 72 % and raising the pooled R^2 from 0.149 to 0.936. The same relation fitted independently at every wavelength, with 722 coefficients instead of four, reaches a pooled R^2 of only 0.825 — more freedom, worse generalisation. With nine training pairs, constraining the model is what makes it transfer, which is the same lesson the standardisation literature draws from far larger transfer sets [16, 18].

The practical consequence is that an instrument costing roughly €166 in components can be taken into the field and produce absorbance values that mean approximately the same thing as a benchtop instrument’s — the precondition for pooling measurements across instruments and sites. The accuracy is not that of a laboratory spectrophotometer, and the limitations above bound what may be claimed; closing the largest of them, the wavelength mapping and the stray-light floor, is what would let one correction be fitted once and reused across units.

ACKNOWLEDGMENT
DATA AND CODE AVAILABILITY

The enclosure designs, the instrument firmware, the calibration and analysis tools and the scripts that generate every figure in this letter are available as supplementary material at <https://github.com/alexmanuel27/MonoSpectro>, under the MIT licence.

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